

VEDANT THAKRE

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SUMMARY

- Engineer building autonomous decision systems and the data infrastructure under them; B.Tech in Aerospace Engineering with a minor in Machine Intelligence and Data Science; built Tark, a causal decision engine for consumer brands; previously the founding data engineer at Epik

EXPERIENCE

Tark — Founder & Engineer

Oct 2025 – Jul 2026

- Consumer brand data sits split across ad platforms, storefront, payments and logistics with no way to trace outcomes back to causes; built Tark, a causal decision engine that closes that gap
- Architected the decision core: an Observe/Think/Act/Measure state machine that calls an LLM only when a confidence gate trips, in under 5% of cycles, paired with a causal model that estimates an action's effect before it is taken
- Built the execution chain to fail closed: circuit breakers, hard caps and idempotent claiming control every action, canary testing and risk limits gate what ships, and measurement flags results it cannot confidently verify

Epik — Founding Data Engineer

Jul 2025 – Sep 2025

- Built 7 ingestion pipelines on BigQuery processing 2.3M records daily from production systems into a warehouse the team could query directly, cutting reporting time to minutes
- Designed and shipped the real-time pricing and supply-demand pipelines in four weeks, replacing manual estimates with live numbers

Independent Consulting — Data Engineering & Deal Sourcing

May 2024 – Jun 2025

- Delivered data engineering and automation projects for investment firms, consulting and healthcare clients, scoping each engagement directly with the client, then building and handing over the system
- Built a deal-sourcing pipeline for an investment firm: crawlers processed 4,800+ companies across 9 registry and filing sources, with an LLM layer scoring each target against the fund's thesis

DevCom, IIT Bombay — Project Lead, Position of Responsibility

Dec 2022 – Mar 2024

- Elected to lead DevCom, IIT Bombay's 21-developer team owning every core digital product on campus, from InstiApp to internal tools; ran the roadmap, release cycle, hiring and annual budget
- Overhauled InstiApp with a new design system and sustained performance work, driving a 10% MAU increase across 5,000+ active students; earlier built the achievement-verification and student-discussion-forum modules

SELECTED PROJECTS

- **Rural Patient Screening (BIRAC-funded):** preliminary diagnostic screening for villages in Maharashtra, reading hospital diagnosis slips with an LLM; fine-tuned LLaMA on real slip data into a compact, domain-specific small language model
- **InstiApp:** core contributor and later lead of IIT Bombay's campus app, used daily by 5,000+ students; shipped digital student IDs, meal access, achievement verification and discussion forums
- **Intelligent Trading Agent:** reinforcement-learning agents (DQN variants) trained and backtested on real market data for sequential decision-making under uncertainty
- **On-Device Drone Autonomy:** obstacle avoidance, path planning and real-time image stitching running entirely on the drone's onboard compute rather than a ground station

TECHNICAL SKILLS

- **Languages & Data:** Python, TypeScript/JavaScript, SQL, Bash, Dart; Trino, BigQuery, Snowflake, PostgreSQL, Firestore, Redis; warehouse and star-schema design, ETL/ELT, event-driven pipelines, semantic layers, query optimisation
- **Cloud & AI:** GCP (Cloud Functions, Cloud Run, Cloud Scheduler, Pub/Sub, BigQuery, Compute Engine), Firebase; LLM agent systems, text-to-SQL, MCP servers, machine learning pipelines

EDUCATION & AWARDS

- **Indian Institute of Technology Bombay: B.Tech, Aerospace Engineering; Minor in Machine Intelligence and Data Science;** coursework in machine learning, data analysis, optimisation, adaptive and learning control, control systems and flight dynamics
- Winner of the institute-wide Game Dev Hackathon at IIT Bombay, which led directly to my election as lead of the DevCom team